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Are companies better off with AI? The effect of AI service failure events on firm value

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Abstract

Purpose – As more firms adopted AI-related services in recent years, AI service failures have increased. However, the potential costs of AI implementation are not well understood, especially the effect of AI service failure events. This study examines the influences of AI service failure events, including their industry, size, timing, and type, on firm value.

Design/methodology/approach – This study will conduct an event study of 120 AI service failure events in listed companies to evaluate the costs of such events.

Findings – First, AI service failure events have a negative impact on the firm value. Second, small firms experience more share price declines due to AI service failure events than large firms. Third, AI service failure events in more recent years have a more intensively negative impact than those in more distant years. Finally, we identify different types of AI service failure and find that there are order effects on firm value across the service failure event types: accuracy > safety > privacy > fairness.

Originality/value – First, this study is the initial effort to empirically examine market reactions to AI service failure events using the event study method. Second, this study comprehensively considers the effect of contextual influencing factors, including industry type, firm size and event year. Third, this study improves the understanding of AI service failure by proposing a novel classification and disclosing the detailed impacts of different event types, which provides valuable guidance for managers and developers.

Keywords Artificial intelligence (AI), Service failure, Event study, Firm value, Accuracy, Safety

Paper type Research paper

1. Introduction

In recent years, artificial intelligence (AI) has boomed and become the most compelling technology that empowers businesses (Lee *et al.*, 2018). Due to AI's excellent capability to analyze and learn from big data, AI-related applications are fundamentally reshaping business processes and greatly promoting the productivity of firms (Pwc, 2019). The implementation of AI by companies is expected to spur more competitive edges and bring higher revenues (Lui *et al.*, 2022). More and more firms believe in the benefits of AI. For example, more than 80% of Fortune 500 CEOs consider AI extremely important to their companies' future (Murry, 2017). As a result, there is an increasing number of firms that are enhancing their investment in AI and introducing AI-related applications to their front- and back-office processes to provide services to users and employees (Wamba-Taguimdje *et al.*, 2020).

However, without an understanding of the costs and drawbacks, it remains a question whether companies made the right decision to implement AI. In practice, AI-related service failures are inevitable because of the technology's imperfect capabilities and the uncertainty of the real world (Mahmood *et al.*, 2022; Pavone *et al.*, 2023). AI service failure events have become increasingly common. For example, in March 2023, UnitedHealthcare allegedly used a faulty AI algorithm to override doctors' recommendations and deny health coverage, leading

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to premature discharges from care facilities and substantial out-of-pocket expenses for patients (Ross and Herman, 2023). Service robots, a popular application of AI, are also frequently reported as clumsy due to preposterous expressions and incapability to communicate with customers for several rounds (Filieri *et al.*, 2022). For companies, such AI service failure events can be a serious problem that reduces firm value and public trust. Therefore, the impact of AI on businesses may not always be as positive as expected. In this context, evaluating the effect of AI service failure events is vital to measuring the potential cost and the feasibility of AI implementation.

The extant literature provides an efficient way of evaluating the potential costs of specific events by measuring the changes in a firm's financial value. Using the event study method, a significant number of studies on firm value examined the effects of technology-related events, such as IT investment, information security, and data breach (Dehning *et al.*, 2005; Gwebu *et al.*, 2018; Yayla and Hu, 2011). In particular, scholars have begun to focus on the impacts of AI-related events (e.g. AI investment and implementation of service robots) in recent years (Lui *et al.*, 2022), leading to an emerging interest in AI service failure events. However, previous studies only investigated AI failure events' impact at the user level and lacked empirical evidence on the magnitude of the influence on firm value. Additionally, the type of service failures has been proven to be a key factor influencing service outcomes (Choi *et al.*, 2021; Song *et al.*, 2023). Even though it is possible to classify service failures based on service phase and service function, such methods ignore users' requirements for service results. In the current research, we adopt prior classifications of service failures into accuracy service failures, safety service failures, fairness service failures, and privacy service failures (Choraś *et al.*, 2020; European Commission, 2019; Kieslich *et al.*, 2022) and examine the effects of different service failure types. Specifically, this study aims to answer the following research questions:

RQ1. Whether and to what extent do AI service failure events impact firm value?

RQ2. How does this impact differ for different companies and in different event years?

RQ3. Whether and to what extent do different types of AI service failure events (accuracy, safety, fairness, and privacy) affect firm value?

To address these research questions, this study evaluates the impact of AI service failure events on firms' market value using the event study methodology, which could isolate and measure the unique contribution of a specific event to a firm's financial value (Hyman and Mathur, 2005). We identified a data set of 120 relevant AI service failure events from 2010 to 2024 and analyzed market reactions to these AI service failure events.

From a theoretical perspective, we contribute to the understanding of AI service failure events in four ways. First, this study is the initial effort to apply signaling theory to explain the market impact of AI service failure events and empirically evaluate their impact on firm value. Second, this study provides a deeper understanding of AI service failure events' impact by examining three contextual influencing factors: industry type, firm size, and event year. Third, this study broadens the service failure literature by proposing a novel classification of service failure. Fourth, this study enriches enterprise resource management research from the perspective of considering the detailed impacts of different AI service failure event types.

2. Literature review

2.1 AI service and service failure events in firms

The extant literature defines AI as a general term that involves machines or computer systems capable of mimicking human intelligence (Huang and Rust, 2018; Longoni *et al.*, 2019). AI representation includes three main types, namely robot, virtual, and embedded AI (Glikson and Woolley, 2020). Because of the distinct abilities of AI in analyzing, learning from, and making autonomous decisions based on data (Huang *et al.*, 2019), more firms now regard AI as a disruptive technology that improves business performance (Wamba-Taguimdje *et al.*, 2020). In practice, businesses use various AI applications to provide customers with product recommendations, content review, navigation, and other services in the front office and provide employees with resume screening, interviewing, work scheduling, supervision, and other services in the back office (Langer and Landers, 2021; Rana *et al.*, 2022).

Despite the rapid development and excellent capabilities of AI, service failure events in AI-related applications are still inevitable. To better understand the definition of AI service failure events, we clarify the definitions of relevant key terms. First, an event is defined as an important happening that occurs "in a certain place during a particular period of time" and has a certain effect (Weiss and Cropanzano, 1996). Second, a service failure refers to "a degraded state of capability that would cause the service provided by the system to deviate from the desired or normal function" (Brooks, 2017). Service failures include AI service failure and robot service failure due to differences in service providers. For example, robot service failure is defined as "the service contact situation in which the robot service does not satisfy the customer's expectations," while AI service failure refers to "the situation in which the service provided by various forms of AI applications deviate from the customer's expectations" (Leo and Huh, 2020). These definitions emphasize the undesirable outcomes of service delivery and the mismatch between the outcomes and expectations. Finally, the definition of "AI service failure events" combines the definitions of event and AI service failure. A related definition, named "AI incident," is proposed by McGregor as a situation in which the service performed by AI systems nearly or actually generates real-world harm (McGregor, 2021). Similarly, this study focuses on "AI service failure events" and defines it as a real event or phenomenon that takes place at a certain time and space in firms and in which the service provided by various AI applications creates undesirable service outcomes or real-world harm.

Several studies in the AI service failure literature have proposed taxonomies of service failures and the impacts of different types of service failures on service consequences. However, previous studies roughly divided service failures into two categories from the perspective of service stages and service functions, such as result failures and process failures (Choi *et al.*, 2021) and functional failures and nonfunctional failures (Song *et al.*, 2023). Few studies have produced a more nuanced classification of AI service failures, especially considering the most important issues that AI-related applications should satisfy (Choraś *et al.*, 2020; European Commission, 2019; Kieslich *et al.*, 2022). For example, AI should satisfy seven requirements, including explainability, fairness, security, accountability, accuracy, privacy, and machine autonomy (Kieslich *et al.*, 2022). Specifically, accuracy is the most basic requirement of users for service delivery and is considered one of the core dimensions to measure service outcomes, namely service quality (Parasuraman *et al.*, 1991). Safety issues, particularly the adoption of AI technologies such as autonomous driving, may result in misdiagnoses of medical conditions, self-driving vehicles losing control, and death (Jörling *et al.*, 2019; Yampolskiy, 2019). Service failure events, such as over-recommendation and collection and storage of large amounts of personal private data, may have undesirable consequences because the service results violate the privacy of individuals (Chi *et al.*, 2020). In addition, AI systems learn from programmed data sets and generate biased outcomes, resulting in fairness issues (Chi *et al.*, 2020). Thus, accuracy, safety, fairness, and privacy issues are related to service outcomes and represent the basic functional requirements. In the process of service interaction, users expect prompt, adaptable, transparent, and accountable services. However, the black-box nature of AI creates difficulties for users in understanding the reasoning process of AI (Chi *et al.*, 2020). The lack of appropriate regulation makes it difficult for users to understand who should be responsible for the negative results, which causes the problem of fuzzy distribution of rights and responsibilities (Čartolovni *et al.*, 2022). Complex and flexible interaction scenarios put forward higher requirements for the autonomy of AI (Lawless *et al.*, 2019). Thus, explainability, accountability, and machine autonomy issues are related to the service process and represent the experience-related requirements. In the current stage of the rise of AI services, individuals pay more attention to whether the AI service results are satisfactory, but there are no clear and strict requirements for the service process. Meanwhile, the former four aspects can be quantified or have established criteria to determine the occurrence of AI service failure events, while the latter three aspects lack uniform rules or standards to assess whether the AI service has failed or not. Accordingly, we focus on four types of AI service failure events from the perspective of AI service result requirements, namely: accuracy service failure event, safety service failure event, fairness service failure event, and privacy service failure event. It should be noted that the categories of AI service failure events are not mutually exclusive since failed service outcomes may involve more than one dimension. We identify the type of an AI service failure event by capturing the most significantly affected dimension of the event. The definitions and examples of these service failure events are shown in Table 1.

Previous studies of AI service failures have focused on their impact on individual users. For example, Lee *et al*. found that robot service failures have severe impacts on consumer satisfaction with the service and perceptions toward the robot (Lee *et al*., 2010). Sun *et al*. showed that AI personal assistant (AIPA) service failures do not stop consumers from using AIPA directly but decrease their continuous use willingness of the failed functionality (Sun *et al*., 2022). Chuang *et al*. (2012) identified two types of service failures – core service failures and interactive service failures – and found that the type of service failure determines customers' evaluation of service recovery efforts. However, the impact of AI service failures at the firm level is not well understood. Thus, it is crucial to explore the effect of AI service failure, especially the influences of different types of AI service failures, on firms.

2.2 The impact of technology-related events on firm value

As technology evolves, prior studies have sought to assess the impact of technology-related events on firm performance (Tayaksi *et al*., 2022; Wamba-Taguimdje *et al*., 2020) Given the difficulty in estimating intangible organizational performance costs, such as loss of competitive advantage and loss of investor confidence, studying the financial impact of technology-related events is critical to understand the potential and actual consequences

Table 1. Definitions and examples of AI service failure events

AI service failure event type	Definition	Examples
Accuracy service failure event	A real event or phenomenon in which the service provided by AI is not able to meet consumers' service needs of accuracy or a major accuracy error or defect is in the results of the service	AI's false or missing results in the area of recommendation, decision-making, etc.
Safety service failure event	A real event or phenomenon in which the service provided by AI is not able to meet consumers' service needs of safety or an unsafe result for life and property is generated by the service	Autopilot malfunction, personal account face recognition failure, etc.
Fairness service failure event	A real event or phenomenon in which the service provided by AI is not able to meet consumers' service needs of fairness or an unfair result is generated by the service	Discriminatory outcomes (race, gender, etc.)
Privacy service failure event	A real event or phenomenon in which the service provided by AI is not able to meet consumers' service needs of privacy or a privacy invasion result is generated by the service	Collection, use, and dissemination of personal information without permission

Source(s): Table created by authors

of these events. The event study method has been widely used to investigate the stock market's reactions to technology-related events and the impacts of these events on firm value (Ebrahimi and Eshghi, 2022). The foundation of event study is the efficient market hypothesis, which argues that a firm's stock price reflects its true value determined by all relevant public information and that a firm's stock price fluctuates instantly to reflect the influence of new information or a specific event (Fama, 1970). Thus, event study is a well-established method to objectively assess how investors evaluate technology-related events by estimating the abnormal change in stock returns.

Studies on firm value have explored the impact of various types of technology-related events, including IT investment, information security events, and data breach events, among others (Dehning *et al.*, 2005; Gwebu *et al.*, 2018; Yayla and Hu, 2011). Meanwhile, there is an emerging interest in the impact of AI-related events. Lui *et al.* explored the effect of AI investment on the firms' market value (Lui *et al.*, 2022). Fotheringham and Wiles examined the relationship between the implementation of AI customer service chatbots and abnormal stock returns (Fotheringham and Wiles, 2023). However, less is known about the impact of AI service failure events which are more complex compared with previously examined AI events. Therefore, this study focuses on the impact of AI service failure events on firm value.

2.3 Signaling theory

Signaling theory was proposed by Spence (1978) and argues that signalers can convey observable signals and disclose information about unobservable factors to the signal receivers to reduce information asymmetry (Chiang *et al.*, 2023). In the stock market, signals can provide information about firms' capabilities and intentions to the market (Connelly *et al.*, 2011). For example, information senders (firms, media, etc.) convey information about corporate value, profitability, business strategy, and other aspects to information receivers (investors, creditors, consumers, and other stakeholders) through specific means (e.g. financial reports, dividend policies, stock policies, and news reports) (Bergh *et al.*, 2014; Connelly *et al.*, 2011).

The extant literature has applied signaling theory to explain market reactions to a variety of information. The signals available through announcements can be either positive or negative. On the one hand, previous studies have found many positive signals had positive impacts on firm value, including acquisitions (Tao-Schuchardt *et al.*, 2023), brand management efficiency (Rahman *et al.*, 2018), and open knowledge disclosure (Liu *et al.*, 2024). In recent years, with the growth of research interest in AI, signaling theory has been extended to the study of market reactions to AI adoption announcements (Nishant *et al.*, 2023). On the other hand, there were concerns about negative signals, and data breaches were regarded by stakeholders as negative signals in the IT context (Chen *et al.*, 2023). However, there is little literature on the effects of negative signals related to AI. Thus, it is necessary to use signaling theory to explore the signals that may have negative impacts on firm value in the context of AI service, namely AI service failure events.

Another research stream of signaling theory investigates the impact of service failure on a series of dependent variables. Numerous studies have shown that service failure can be seen as a signal of poor service quality (Zhao *et al.*, 2024) and has negative consequences on individuals, including negative word of mouth (Jean Harrison-Walker, 2012) and switching intentions and behaviors (Zhao *et al.*, 2023). Among the limited studies that explored the impact of service failure on firms, one has revealed that data breaches, as a form of service failure, convey information about the vulnerability of information systems and have a negative impact on firm value (Chen *et al.*, 2023). Similarly, in the context of AI, service failure events imply that the company has inadequate AI capabilities (Perifanis and Kitsios, 2023) to deliver accurate and quality services, thus conveying a negative signal. Therefore, signaling theory provides a useful theoretical perspective to explore market reactions to AI service failure events. Considering that the interpretation of a signal by receivers such as shareholders determines their responses to the signal (Bergh *et al.*, 2014; Connelly *et al.*, 2011), examining how market reactions differ based on firm type, event year, and event type will allow us to obtain a more nuanced understanding of AI service failure events as a signal.

3. Hypotheses development

3.1 Effect of AI service failure events on firm value

In an efficient market, stock prices generally reflect investors' expectations of and confidence in future firm performance (Bose and Pal, 2012). A firm's market value fluctuates instantaneously in response to investors' evaluation of specific events' costs and benefits (Shiller, 2000). The market value will decrease if the costs and risks brought by focused events are greater than the benefits and vice versa. In previous studies, security breach events are considered a negative phenomenon since they may cause undesirable harm to customers' trust, satisfaction, and firm reputation (Campbell *et al.*, 2003; Modi *et al.*, 2015). Most research results revealed that security breach events have a significantly negative impact on firm value (Ebrahimi and Eshghi, 2022; Spanos and Angelis, 2016).

Similarly, from the perspective of investors, AI service failure events are a negative signal in the stock market and can be regarded as a potential threat to a firm's future continuity and success. First, AI service failure itself will directly increase the financial cost due to possible lawsuits, compensation to customers, and investment in service repair, thus influencing future cash flow. Second, AI service failure events can induce users' negative evaluation of and satisfaction with the services and damage word-of-mouth and firm reputation, resulting in a decrease in future profits. Third, a company's initial decision to implement AI may be a contentious issue for users and investors if they have doubts about AI service quality and are concerned about privacy, security, fairness, and other issues of AI services. The occurrence of an AI service failure event verifies their concerns, which can easily trigger doubts about the company's decision capability. Considering all these tangible and intangible costs of AI service failure events, investors have pessimistic expectations about the firm's profitability in the short term, ultimately generating a negative market reaction. Thus, we posit that:

H1. An AI service failure event has a negative impact on a firm's market value.

3.2 Effect of industry type on firm value

As a type of technology-related event, AI service failure events have different meanings for firms in different industry types. The same signal can be interpreted in different ways since events occur to different types of firms. Firms in the technology industry are expected to provide technology-related services with higher quality and fewer errors since they have more technical knowledge and capabilities in advanced technologies than non-technology firms (Yayla and Hu, 2011). When negative technology-related events take place in technology firms, such violations of expectancy will lead to more negative outcomes. For example, previous research has shown that firm value in the technology industry is more vulnerable to information security breaches (Pirounias *et al.*, 2014; Yayla and Hu, 2011). Similarly, when a technology firm fails to provide high-quality AI services, investors might lose high expectations and confidence in the firm's future, resulting in a much stronger negative impact of the service failure event compared with the same event in a non-technology firm. In addition, technology firms themselves are more reliant on AI services to improve their business processes, which exacerbates the perception of firm profitability. Thus, we posit that:

H2. An AI service failure event will have a stronger negative impact on the market value of a technology firm than that of a non-technology firm.

3.3 Effect of firm size on firm value

Firm size is an important factor that may impact market reactions to AI service failure events due to different sensitivities to the same signal. Many studies have examined the role of firm size in market reactions to negative events but found mixed results. Some scholars believe that large companies are more vulnerable to negative events such as data breaches due to higher volatility (Yayla and Hu, 2011). However, the more mainstream view is that larger firms tend to experience lower abnormal returns since they have multiple revenue streams, a diversified product portfolio, and an established brand name (Cavusoglu *et al.*, 2004). As a result, large firms can better overcome the consequences of such negative events than small firms (Ebrahimi and Eshghi, 2022). It is well documented in the information systems literature that the market value of larger firms is less adversely impacted by announcements of negative events, such as security breaches (Ebrahimi and Eshghi, 2022) and IT failures (Bharadwaj and Keil, 2001). Similarly, AI service failure events can be viewed as negative economic shocks and large firms are likely to handle these shocks better than small firms, resulting in a decreased negative market reaction. Thus, we posit that:

H3. An AI service failure event will have a stronger negative impact on the market value of a small firm than that of a large firm.

3.4 Effect of event year on firm value

Prior studies have shown that the time period of an event is important when studying stock market reactions to specific events. As a result, it may influence how stakeholders interpret the signal conveyed by AI service failure events. The change in impact magnitude reflects investors' attention and sensitivity to the event in different time periods. However, there are no conclusive results in previous studies on how the effect of security breach events evolves in different time periods. On one hand, Cavusoglu *et al*. found that the stock market reacts more intensely to recent attacks (Cavusoglu *et al.*, 2004). On the other hand, others indicated that security breach events that happened in earlier years brought more severe outcomes on stock values (Gordon *et al.*, 2011; Yayla and Hu, 2011). One probable reason is that users' concern changes as the technology matures. In the early days of a technology, little is known about its uncertainties and risks. With the increase in negative events, investors gradually pay more attention to negative events and react more harshly. In contrast, at a later stage of a technology's development, the abundance of negative event announcements may not trigger investors' attention or negative reaction. In the context of AI service, we posit that it is in the early stage of AI development where people are trying to understand the emerging technology. As more firms implement AI services, people are increasingly focused on AI service quality and AI-related events and may discuss the rationality of AI applications more frequently. These voices of doubt indicate that customers are becoming more aware of AI service risks and less tolerant of AI service failure events, resulting in investors' concerns about business performance. Accordingly, AI service failure events in more recent years tend to impact the market value of firms more intensively. Thus, we hypothesize that:

H4. An AI service failure event that occurred in recent years has a stronger negative impact on the market value of a firm than one that occurred in earlier years.

3.5 Effect of AI service failure event type on firm value

Previous research has highlighted the importance of event type on stock value (Ebrahimi and Eshghi, 2022). In the context of AI service, firms may experience different types of AI service failure events including accuracy service failure, safety service failure, fairness service failure, and privacy service failure. The effect magnitude of different types of AI service failure events depends on the degree of users' requirements urgency and the size of estimated costs, which convey signals of varying strength to stakeholders in assessing firms' AI capabilities.

From the perspective of user requirements, the accuracy of AI service results should be the most basic and important requirement of users. Prior research indicates that users have initial performance expectations for AI services, and failing to meet these expectations after a user interacts with AI will have a significant negative impact on trust (Dietvorst *et al.*, 2015; Glikson and Woolley, 2020). In addition, there is the additional requirement of trustworthy AI to satisfy users' expectations, including safety, fairness, and privacy (Choraś *et al.*, 2020; Kieslich *et al.*, 2022). Safety issues are related to the safety of users' lives and property. Fairness issues involve prejudice and discrimination against particular groups, which is a key ethical principle for AI service (Starke *et al.*, 2022). Privacy issues are an important ethical risk for users' sensitive personal information. However, empirical evidence shows that humans have become less sensitive to privacy events in recent years (Yayla and Hu, 2011). Users may regard privacy as a common risk for using a technology, thus having a lower requirement for privacy than fairness. Security issues are related to users' vital interests and may lead to more serious consequences than fairness or privacy issues.

From the perspective of estimated costs, accuracy and safety service failures are objective and more likely to lead to both tangible financial costs and intangible user trust and corporate image costs. In contrast, fairness and privacy service failures are subjective moral issues that come with higher intangible costs on user satisfaction. Moreover, fairness service failures are often regarded as unintentional results due to biased data sets (Suresh and Guttag, 2021). Thus, investors may be more tolerant of fairness service failures.

Overall, we posit that the negative impacts of different AI service failure types on firm market value are different:

H5. The negative impacts of accuracy service failures, safety service failures, fairness service failures, and privacy service failures on firm value are different.

4. Method

We empirically investigated the effect of AI service failure events on firm value in the stock market using the event study methodology. The event study method is built based on the efficient market hypothesis, which argues that the change in a firm's stock price surrounding a particular point reflects the overall impact of a particular event or relevant information cues (Fama, 1970). Event studies have been widely used in prior research because they can generate reliable results even in imperfect conditions, capturing the instantaneous effect of focal events and ignoring the effects of other concomitant factors (Bose and Pal, 2012; Sorescu *et al.*, 2017). Accordingly, event studies provide a useful approach for us to capture the financial impacts of firms' AI service failure events in the short term.

4.1 Data collection and sample

The AI service failure event sample was collected from related public open database such as AI Incident Database (AIID) and AI, Algorithmic, and Automation Incidents and Controversies (AIAAIC) Repository. AIID and AIAAIC Repository both systematically collect AI-related service failure event information from user submissions and provide website links of news announcements related to the AI service failure event which includes key news sources such as Wall Street Journal and New York Times. They have been used in prior AI research (Nasim *et al.*, 2022; Zhan *et al.*, 2023). We combined them to generate a more comprehensive AI service failure event dataset. We checked all news announcements links of AI service failure events and used the earliest release date of news as the event date. If the news was released after the market closed, the event date was postponed to the next day.

The search from AIID and AIAAIC yield an initial AI service failure event data set of 1,385 events. Following previous research, we filtered out irrelevant or duplicate events as follows. First, we removed duplicate events and preserved only events involving companies listed on the public stock exchanges NYSE and NASDAQ, resulting in 451 events. Second, because we used the range of [−130, −11] days before the event as the estimation period of stock prices, events involving companies that were not listed during the estimation period were excluded. Events were also removed from the sample if the time interval between two events occurring in the same firm was less than 130 trading days, resulting in a sample of 129 events. Third, to avoid the effect of confounding events, events with potential confounding news, including earnings announcements/dividends, acquisitions/mergers, and top management changes, within the event window were excluded. For example, if an acquisition announcement occurred on the fourth day after the event date ($t = 0$), the data point was included in the analyses with event windows of [0, 1] and [0, 3], but dropped from the analysis with event window [0, 5]. As a result, our sample size varies across different event windows. Following these procedures, a total of 120 AI service failure events from 75 firms were identified between 2010 and 2024. The data processing process is summarized in Figure 1, and the distribution of the AI service failure events over time and across industries is shown in Tables 2 and 3. The extant literature operationalizes firm size in various ways including the number of employees (Foerderer and Schuetz, 2022), market capitalization (Tripathi and Mukhopadhyay, 2020), and total assets and total sales (Dang *et al.*, 2018). This study measures the firm size by whether a firm is in the Fortune Global 500 List (based on sales and profits) or the PwC Market Capitalization Top 100 List (by market capitalization) in the year of the event occurrence. A firm on either list was coded as large and those not are coded as small. The distribution of the events by firm size is shown in Table 4, and the detailed information of firms in our sample is shown in the Appendix.

Three research assistants coded the AI failure service event type. Before coding started, they read definitions and examples of the four service failure event categories. Next, they manually reviewed the description of each event and independently coded the category to which the service failure event belonged. Considering that some events might belong to more than one category, the research assistants coded the categories of events in a ranked manner so

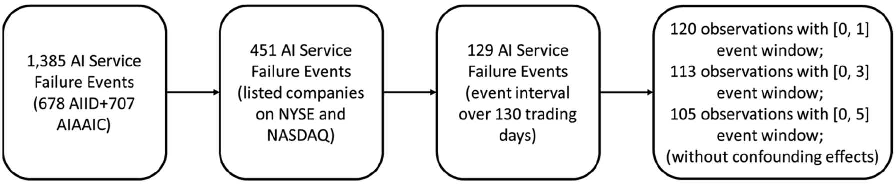

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Figure 1. The data processing process

Table 2. Distribution of events by year

Year	Number	Year	Number
2009	1	2017	8
2010	1	2018	8
2011	1	2019	14
2012	2	2020	9
2013	1	2021	14
2014	5	2022	20
2015	2	2023	25
2016	5	2024	4
Total	120		

Source(s): Table created by authors

Table 3. Distribution of events by industry

Industry type	Count
Technology firms	82
Non-technology firms	38

Source(s): Table created by authors

Table 4. Distribution of events by firm size

Firm size	Count
Large firms	70
Small firms	50

Source(s): Table created by authors

that they could identify the greatest impact aspect of service failure results and the second influencing aspect. Then, the ranking results of several coders were integrated and each AI failure service event was classified into a category that it most likely belonged to. The coding process of AI failure service event type obtained consistent results with an inter-coder reliability of over 80%, and the disagreements were resolved by following the coding results of the majority and further discussions among the coders. Table 5 provides examples of the sample coding results, and the distribution of the AI service failure events across types is shown in Table 6.

4.2 Estimation method

In standard event studies, the first step of analysis is to calculate the normal (expected) stock return for the situation had the event under consideration not occurred. By applying a market model based on the capital asset pricing model (CAPM), the normal (expected) stock return can be presented as:

$$R_{it} = \alpha_i + \beta_i R_{mt} + \varepsilon_{it}, \tag{1}$$

Table 5. Examples of sample coding results

Firm	Market	Event date	Event description	Event type
CI	NASDAQ	2023/3/27	Cigna's AI algorithms did not examine hundreds of thousands of patient claims as California law requires and incorrectly denied them	Accuracy
TSLA	NYSE	2016/5/26	A Tesla Model S crashed into a van while on autopilot	Safety
GOOGL	NASDAQ	2013/2/4	Google Adsense AD selection was reported to produce sexist and racist results	Fairness
MCD	NYSE	2021/6/7	McDonald's was charged in a lawsuit in Chicago with collecting and processing voice data to predict customer information without user consent in its AI drive-thru chatbot	Privacy

Source(s): Table created by authors

Table 6. Distribution of events by event type

AI service failure event type	Count
Accuracy service failures	39
Safety service failures	26
Privacy service failures	23
Fairness service failures	32

Source(s): Table created by authors

where R_{it} is the normal (expected) stock return of the *i*th firm on day *t*; R_{mt} is the return of the market index on day *t* (Nasdaq Composite Index and NYSE Composite Index); α_i and β_i are the intercept and the slope of the market model for the *i*th firm, respectively; ε_{it} is the error term.

We distinguish between the AI service failure event's date of occurrence and the event date used in our analysis. From the investors' perspective, the date of a firm-specific AI service failure event is the day when such an event becomes public knowledge through media announcements or via business news, which is not necessarily the date when the event occurred. In this study, we used the announcement event date for each AI service failure event. Meanwhile, we set the estimation period to 120 days, from $t = -130$ days to $t = -11$ days, where $t = 0$ day represents the event date.

Next, we calculated the abnormal returns for each day, which is the difference between the actual and the expected normal return of the stock of firm *i* on day t (R_{it} here denotes the actual daily returns):

$$AR_{it} = R_{it} - (\alpha_i + \beta_i R_{mt}). \tag{2}$$

Then, the cumulative abnormal returns (CAR) can be calculated by aggregating all abnormal returns for a firm within the event window, which measures the overall impact of the event (Campbell *et al.*, 2003). The CAR of firm *i* during the event window is:

$$CAR_i = \sum_{t_1}^{t_2} AR_{it}, \tag{3}$$

where t_1 and t_2 represent the start and end dates of the event window in which abnormal returns were cumulated.

For all N firms in the sample, the average CAR was calculated as:

$$\mathrm{CAR} = \frac{1}{N} \sum_{j=1}^{N} CAR_i. \tag{4}$$

If the calculated CARs are significantly different from zero, then there are abnormal returns due to the events. That is, the changes in the stock prices of the firms during the event window are caused by the underlying events.

5. Results

5.1 Analysis of the full sample

Using daily stock prices of the firms, the market parameters of Equation (1) were estimated for each sample firm and then ARs and CARs were calculated using Equations (2–4). The results for the full sample are presented in Table 7.

Table 7. Results from event analyses using the full sample

Event window	Sample size	Mean CAR	t-value
[0, 1]	120	−1.37%	−3.662***
[0, 3]	113	−2.25%	−4.066***
[0, 5]	105	−2.00%	−2.770***
[−1, 1]	120	−1.32%	−2.886***
[−1, 3]	113	−2.20%	−3.569***
[−1, 5]	105	−2.00%	−2.559**

Note(s): ***$p < 0.01$; **$p < 0.05$; *$p < 0.10$
Source(s): Table created by authors

For the event window [0, 1], the mean CAR was −1.37% ($p < 0.01$) for all firms, indicating that between the AI service failure event day and the day after the announcement, the portfolio of the firms in the sample experienced an average 1.37% decrease in their stock prices compared with their market valuations based on the CAPM model. Compared with other studies that investigated the effects of security breach or IT disruption announcements using the same event window, the magnitude of AI service failure events in our study was considerably stronger (e.g. Yayla and Hu (2011) reported −0.92%, Gwebu *et al*. (2018) reported–0.41%) (Gwebu *et al.*, 2018; Yayla and Hu, 2011).

We further analyzed this sample with [0,3] and [0, 5] windows to investigate the longevity of the negative effect of the announcements. The mean CAR for the [0, 3] window was −2.25%, significant at the $p < 0.01$ level. The mean CAR for the [0, 5] window was −2.00%, significant at the $p < 0.01$ level. These effects were both stronger than the previous result of −1.37% for the [0, 1] window, suggesting that a delayed reaction to the news by some investors might have occurred. Thus, a longer time frame was also analyzed to verify that these effects were not just for the short term. The mean CAR for the [0,5] window was −2.00%, slightly smaller than that for the [0,3] window. This result could be interpreted as the market valuations of the firms were indeed recovering from the initial shocks during the 5-day period following the announcements. The findings showed that there was a statistically significant negative stock market reaction to the announcements or news of firm-specific AI service failure events, and that this negative effect varied depending on the length of the event window.

5.2 Analysis of effect by industry type

To test H2, we divided the full sample into two sub-samples: technology vs non-technology firms. Table 8 presents the results from event analyses based on the industry type. For

Table 8. Event analysis results by industry type

Event window	Technology firms Sample size	Mean CAR	t-value	Non-technology firms Sample size	Mean CAR	t-value	Difference in mean CAR	Mann-Whitney U-test
[0, 1]	82	−1.38%	−2.681***	38	−1.34%	−3.280***	−0.04%	−1.309*
[0, 3]	77	−2.41%	−3.156***	36	−1.91%	−3.149***	−0.50%	−0.253
[0, 5]	71	−2.14%	−2.118**	34	−1.80%	−2.342**	−0.33%	−0.240
[−1, 1]	82	−1.41%	−2.233**	38	−1.13%	−2.260**	−0.27%	−0.401
[−1, 3]	77	−2.43%	−2.856***	36	−1.71%	−2.533**	−0.72%	−0.327
[−1, 5]	71	−2.22%	−2.039**	34	−1.61%	−1.932*	−0.61%	−0.281

Note(s): ***$p < 0.01$; **$p < 0.05$; *$p < 0.10$. Mann-Whitney U tests were one-tailed
Source(s): Table created by authors

technology firms, the mean CARs for the [0, 1], [0, 3] and [0, 5] windows were −1.38% ($p < 0.01$), −2.41% ($p < 0.01$), and −2.14% ($p < 0.05$), respectively. These results suggest that there was a significant drop in market value for technology firms after the events. As for non-technology firms, the mean CARs for [0, 1], [0, 3] and [0, 5] windows were all significantly negative. These results indicate that AI service failure events had a significant negative impact on market value for non-technology firms.

We further compared the mean CARs of the industry sub-samples. Considering that the sample size was small and the data were not normally distributed, we conducted the Mann-Whitney *U*-test, a nonparametric test equivalent to the independent sample *t*-test. As shown in Table 8, results of the Mann-Whitney *U*-test revealed that the difference between the mean CARs of the two sub-samples was statistically significant at the 0.1 level for the [0, 1] event window, partially supporting H2. Thus, technology firms experienced a stronger negative drop in their market value than non-technology firms the first day after the AI failure event announcements.

5.3 Analysis of effect by firm size

To test H3, we divided the full sample into two sub-samples: large vs small firms. As shown in Table 9, for small firms, the mean CARs for the [0, 1], [0, 3], [−1, 1] and [−1,3] windows were −1.93% ($p < 0.05$), −3.08% ($p < 0.05$), −2.12% ($p < 0.05$) and −3.29% ($p < 0.05$), respectively. As for large firms, the mean CARs for all windows were significantly negative. These results indicate that AI service failure events had a significant negative impact on market value for both small firms and large firms.

The further comparison analysis on the mean CARs of the sub-samples by firm size is shown in Table 9. The results of the Mann-Whitney *U*-test revealed that the differences between the mean CARs of the two sub-samples were statistically significant at the 0.05 level for the [0, 1], [−1, 1], and [−1, 3] event windows, thus supporting H3. Small firms experienced a stronger negative drop in their market value than large firms after the AI service failure event announcements.

5.4 Analysis of effect by event year

To investigate the effect of event year, the full sample was divided into two sub-samples: events from 2010 to 2019 and events from 2020 to 2024. Since the COVID-19 pandemic promoted the digital transformation of all organizations (Fletcher and Griffiths, 2020), the application of AI in enterprises entered an era of unprecedented growth post the pandemic. As a result, we chose 2019 as the cutoff for the two sub-sample periods. Table 10 summarizes the results of our analyses.

Table 9. Event analysis results by firm size

Event window	Small firms Sample size	Mean CAR	*t*-value	Large firms Sample size	Mean CAR	*t*-value	Difference in mean CAR	Mann-Whitney *U* test
[0, 1]	50	−1.93%	−2.395**	70	−0.96%	−3.360***	−0.97%	−1.810**
[0, 3]	48	−3.08%	−2.608**	65	−1.63%	−3.945***	−1.45%	−1.249
[0, 5]	48	−2.19%	−1.507	57	−1.89%	−3.478***	−0.30%	−0.508
[−1, 1]	50	−2.12%	−2.149**	70	−0.75%	−2.126**	−1.37%	−1.714**
[−1, 3]	48	−3.29%	−2.493**	65	−1.40%	−3.030***	−1.89%	−1.684**
[−1, 5]	48	−2.40%	−1.531	57	−1.70%	−2.891***	−0.7%	−0.669

Note(s): ***$p < 0.01$; **$p < 0.05$; *$p < 0.10$. Mann-Whitney *U* tests were one-tailed
Source(s): Table created by authors

Table 10. Event analysis results by event year

Event window	Event year ≤2019 Sample size	Mean CAR	*t*-value	Event year ≥2020 Sample size	Mean CAR	*t*-value	Difference in mean CAR	Mann-Whitney *U* test
[0, 1]	48	−0.86%	−2.150**	72	−1.70%	−3.019***	−0.84%	−3.198***
[0, 3]	45	−1.25%	−2.105**	68	−2.90%	−3.487***	−1.65%	−3.015***
[0, 5]	43	−1.37%	−1.806*	62	−2.48%	−2.233**	−1.11%	−1.362*
[−1, 1]	48	−0.86%	−1.744*	72	−1.63%	−2.357**	−0.77%	−1.516*
[−1, 3]	45	−1.22%	−1.833*	68	−2.85%	−3.067***	−1.63%	−2.334***
[−1, 5]	43	−1.32%	−1.615	62	−2.50%	−2.086**	−1.18%	−1.056

Note(s): ***$p < 0.01$; **$p < 0.05$; *$p < 0.10$. Mann-Whitney *U* tests were one-tailed
Source(s): Table created by authors

For the 2010–2019 sub-sample, the mean CARs for the [0, 1], [0, 3] and [0, 5] windows were −0.86% ($p < 0.05$), −1.25% ($p < 0.05$), and −1.37% ($p < 0.1$), respectively. For the 2020–2024 subsample, the mean CARs for the [0, 1], [0, 3] and [0, 5] windows were −1.70% ($p < 0.01$), −2.90% ($p < 0.01$), and −2.48% ($p < 0.05$), respectively.

To test H4, we compared the mean CARs of the two sub-samples using the Mann-Whitney *U* test. As shown in Table 10, the one-tailed test results of Mann-Whitney *U* tests revealed that the negative effects of AI failure event announcements in the 2020–2024 group were significantly stronger than those in the 2010–2019 group for the [0, 1], [0, 3], [0, 5], [−1, 1], and [−1, 3] event window. Thus, H4 was supported where AI service failure events announcements between 2020 and 2024 had a stronger negative impact on firms' market value than those between 2010 and 2019.

5.5 Analysis of effect by event type

To analyze the effects of AI service failure event type, the full sample was divided into four sub-samples: accuracy service failure events, safety service failure events, fairness service failure events, and privacy service failure events. Table 11 summarizes the results.

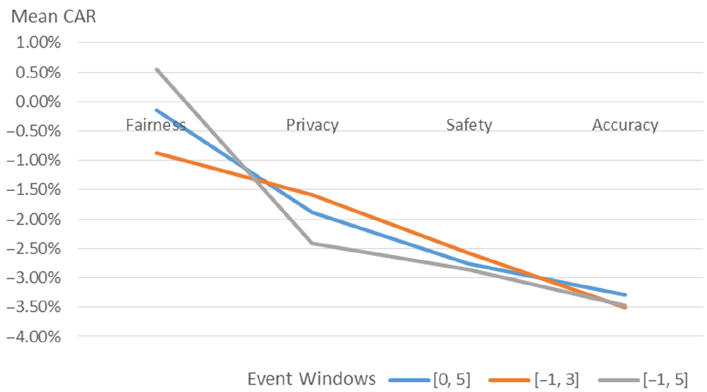
The mean CARs of accuracy service failure events for all event windows were significantly negative. The mean CARs of accuracy service failure events for the [0, 1], [0, 3], and [0, 5] windows were −1.62% ($p < 0.05$), −3.39% ($p < 0.01$), and −3.28% ($p < 0.05$), respectively. For safety service failure events, the mean CARs were −1.46% ($p < 0.05$) for the [0, 1] window, −2.51% for [0, 3], and −2.75% for [0, 5], respectively. For privacy service failure events, the mean CARs was −1.86% ($p < 0.1$) for the [−1, 1] window. As for fairness service failure events, the mean CARs were negative but not statistically significant.

To compare the effects across service failure types, we conducted two non-parametric tests, including the Kruskal-Wallis test and the Jonckheere-Terpstra test. The results of the Kruskal-Wallis test showed a significant difference in mean CARs among the AI service failure event types at the 0.05 level for the [0, 5] window and at the 0.01 level for the [−1, 5] window. The Jonckheere-Terpstra test was used to test whether there were order effects of the mean CAR across the service failure types. AI service failure type was used as an ordinal variable with four values in increasing order: fairness, privacy, safety, and accuracy. The results were statistically significant for the [0, 5], [−1, 3] and [−1, 5] windows, suggesting that the mean CAR decreased as the service failure type value increased. As shown in Figure 2, the mean CAR showed a decreasing tendency for event types. One reason could be that accuracy is users' most basic requirement and inaccurate results will significantly reduce investors' perception of firm capabilities. Therefore, these results provide support for H5 that the negative impacts on firm value across different types of AI service failure events are different.

Table 11. Results of event analyses by event type

Event window	Accuracy service failure Sample size	Mean CAR	*t*-value	Safety service failure Sample size	Mean CAR	*t*-value	Privacy service failure Sample size	Mean CAR	*t*-value	Fairness service failure Sample size	Mean CAR	*t*-value	Kruskal-Wallis test	Jonckheere-Terpstra test
[0, 1]	39	−1.62%	−2.504**	26	−1.46%	−2.072**	23	−1.19%	−1.421	32	−1.11%	−1.332	2.898	−1.538
[0, 3]	36	−3.39%	−3.439***	23	−2.51%	−2.379**	23	−0.92%	−0.775	31	−1.71%	−1.415	3.207	−1.358
[0, 5]	33	−3.28%	−2.504**	22	−2.75%	−2.056**	21	−1.89%	−1.202	29	−0.15%	−0.093	8.828**	−2.865***
[−1, 1]	39	−1.71%	−2.167**	26	−1.55%	−1.794*	23	−1.86%	−1.816*	32	−0.26%	−0.256	3.371	−1.086
[−1, 3]	36	−3.50%	−3.189***	23	−2.58%	−2.187**	23	−1.59%	−1.202	31	−0.87%	−0.642	5.571	−2.321**
[−1, 5]	33	−3.47%	−2.459**	22	−2.86%	−1.975**	21	−2.42%	−1.423	29	0.56%	0.326	12.127***	−3.319***

Note(s): ***$p < 0.01$; **$p < 0.05$; *$p < 0.1$. Kruskal-Wallis and Jonckheere-Terpstra tests were two-tailed
Source(s): Table created by authors



Source(s): Figure created by authors

Figure 2. Mean CARs by event type

5.6 Robustness checks

5.6.1 Placebo test. To cross-validate our results, we conducted a placebo test following Fletcher and Griffiths (2020). We placed the event date of the placebo event at 60 days before the actual AI service failure incident. We adjusted the estimation window accordingly and calculated the mean CAR using identical parameters. As shown in Table 12, the mean CARs for all event windows were nonsignificant. These results indicated that our results were not an artifact of the event dates but were specific to the event announcements. Additionally, given that the COVID-19 pandemic may have had a significant impact on the results, we further estimated market reactions to placebo events between event year sub-samples. The mean CARs for all event windows were not significant in either sub-sample, indicating that the difference among event year groups was not due to the COVID-19 pandemic.

5.6.2 The effect by event year. Considering that the division of the sample around the COVID-19 pandemic introduces potential confounding variables that could disturb the true temporal effects on AI service failure impact, a more granular temporal analysis was necessary. To isolate the influence of earlier vs recent years from pandemic-related effects, we conducted two separate comparison. The influence of earlier years was divided into 2010–2014 vs 2015–2019, and the influence of recent years was divided into 2020–2021 vs 2022–2024. The comparison results were shown in Tables 13 and 14. For earlier years, the effect of AI service failure events on firm value was significantly stronger for subsample (2010–2014)

Table 12. Event analyses results for placebo events

Event window	Full sample Sample size	Mean CAR	*t*-value	Event year ≤2019 Sample size	Mean CAR	*t*-value	Event year ≥2020 Sample size	Mean CAR	*t*-value
[0, 1]	120	0.21%	0.554	52	−0.39%	1.035	68	0.07%	0.113
[0, 3]	120	0.19%	0.351	52	−0.94%	1.747	68	−0.40%	−0.453
[0, 5]	120	−0.10%	−0.155	52	−0.89%	1.344	68	−0.89%	−0.829
[−1, 1]	120	0.07%	0.147	52	−0.05%	−0.100	68	0.16%	0.216
[−1, 3]	120	0.05%	0.078	52	−0.50%	0.829	68	−0.30%	−0.309
[−1, 5]	120	−0.25%	−0.342	52	−0.44%	0.624	68	−0.80%	−0.686

Note(s): ***$p < 0.01$; **$p < 0.05$; *$p < 0.10$
Source(s): Table created by authors

Table 13. Comparison of the influence of earlier years

Event window	Event year ≤2014 Sample size	Mean CAR	*t*-value	2015≤ event year ≤2019 Sample size	Mean CAR	*t*-value	Difference in mean CAR	Mann-Whitney *U* test
[0, 1]	11	−1.38%	−2.163**	37	−0.71%	−1.454	0.66%	*−1.582**
[0, 3]	10	−0.76%	−0.792	35	−1.39%	−1.935*	−0.63%	−0.382
[0, 5]	9	−0.37%	−0.286	34	−1.63%	−1.809*	−1.26%	*−1.400**
[−1, 1]	11	−1.40%	−1.800*	37	−0.70%	−1.161	0.71%	−0.724
[−1, 3]	10	−0.73%	−0.676	35	−1.36%	−1.690*	−0.63%	−0.737
[−1, 5]	9	−0.26%	−0.184	34	−1.60%	−1.644	−1.35%	*−1.316**

Note(s): ***$p < 0.01$; **$p < 0.05$; *$p < 0.10$. Mann-Whitney *U* tests were one-tailed
Source(s): Table created by authors

Table 14. Comparison of the influence of recent years

Event window	2020≤ event year ≤2021 Sample size	Mean CAR	*t*-value	2022≤ event year ≤2024 Sample size	Mean CAR	*t*-value	Difference in mean CAR	Mann-Whitney *U*-test
[0, 1]	23	−1.51%	−1.906*	49	−1.79%	−2.406**	−0.29%	*−1.307**
[0, 3]	22	−3.22%	−2.873***	46	−2.76%	−2.468**	0.46%	−0.944
[0, 5]	18	−2.16%	−1.335	44	−2.62%	−1.829*	−0.46%	−0.465
[−1, 1]	23	−0.94%	−0.966	49	−1.95%	−2.137**	−1.01%	−1.032
[−1, 3]	22	−2.62%	−2.102**	46	−2.97%	−2.377**	−0.35%	−0.079
[−1, 5]	18	−1.84%	−1.065	44	−2.77%	−1.794*	−0.93%	−0.558

Note(s): ***$p < 0.01$; **$p < 0.05$; *$p < 0.10$. Mann-Whitney *U* tests were one-tailed
Source(s): Table created by authors

than subsample (2015–2019) at the event window [0, 1], but significantly stronger for subsample (2015–2019) than subsample (2010–2014) at the event window [0, 5] and [−1, 5], partially supporting H4. While after 2019, the effect of AI service failure events on firm value was significantly stronger for subsample (2022–2024) than subsample (2020–2021) at the event window [0, 1], supporting H4.

5.6.3 Regression analysis. To confirm the robustness and consistency of the findings, we conducted a linear regression analysis to explore the influencing factors of the abnormal returns as suggested by Yayla and Hu (2011). We used mean CAR as the dependent variable and included industry type, firm size, event year, and event type as explanatory variables. Additionally, the magnitude (or severity) of AI service failures' impact on firm value should be considered. However, it is difficult to compare the magnitude (or severity) of the events directly since AI service failure events involve many different types and have different impacts and losses, including economic losses, life injuries, privacy threats, reputation losses, etc. Therefore, we included the impact scope to measure whether the number of people affected by an event is large or small. If the result of an AI service failure event harms the interests of a group, such as women, the elderly, a specific race, etc., then the impact scope of the event is perceived to be greater. For example, the service failure of the smart recruitment algorithm will affect the female group, and the service failure of the insurance algorithm will affect the elderly group. The occurrence of these events will affect a group with similar demographic characteristics, so the number of victims involved is larger. When an event that damages the interests of a group occurs, the public may feel that their interests are damaged due to their relationship with the group, thus they will regard it as a more serious event and take a more drastic reaction. On the contrary, if the result of the AI service failure event damages the interests of individuals, the scope of influence of the event is relatively small. AI service failure events may only happen to certain individuals, such as someone jailed for a wrong facial recognition result or someone injured by an autonomous driving malfunction, which involves only a few people. When an event that damages the interests of individuals occurs, the public may think that the event is accidental and the possibility of it happening to them is low, thus they will regard it as a relatively non-serious event and take a more moderate reaction. We also used a variable to measure whether there was someone injured or killed in the events. These two variables were included as control variables to eliminate the impacts of the magnitude (or severity) of the events. As shown in Table 15, these results were largely consistent with our previous findings, and the significant results in different event windows provide support for H2 to H5.

5.7 Post-hoc analysis: the interaction effects

To advance our understanding of the mixed effects that a simple categorical approach might overlook, we enhanced our analysis by including interaction terms in the regression model.

First, we explored how firm size and industry type jointly affect the impact of AI service failures on firm value. As shown in Table 16, the coefficients of the interaction terms were significantly positive in the event windows of [0,3] and [0,5], suggesting a significant interaction effect between industry type and firm size. When the firm size was small, AI

Table 15. Regression analysis results

	Event window					
	[0,1]	[0,3]	[0,5]	[−1,1]	[−1,3]	[−1,5]
Impact scope	−0.003	−0.011	−0.015	0.002	−0.006	−0.012
	(−0.633)	(−1.365)	(−1.335)	(0.260)	(−0.733)	(−0.957)
I&D (injuries and deaths)	−0.001	0.003	0.011	−0.000	0.003	0.011
	(−0.079)	(0.201)	(0.587)	(−0.003)	(0.235)	(0.564)
Industry type	−0.002	−0.011	−0.013	−0.005	−0.014*	−0.017
	(−0.345)	(−1.354)	(−1.241)	(−0.802)	(−1.704)	(−1.447)
Firm size	0.008*	0.010	−0.003	0.012**	0.014*	0.000
	(1.824)	(1.303)	(−0.275)	(2.055)	(1.754)	(0.036)
Event year	−0.006	−0.013*	−0.008	−0.004	−0.011	−0.006
	(−1.380)	(−1.695)	(−0.726)	(−0.643)	(−1.368)	(−0.528)
Accuracy	−0.004	−0.018*	−0.038***	−0.014*	−0.027***	−0.047***
	(−0.668)	(−1.765)	(−2.793)	(−1.790)	(−2.634)	(−3.198)
Safety	−0.006	−0.019	−0.044**	−0.013	−0.025*	−0.051**
	(−0.723)	(−1.328)	(−2.399)	(−1.233)	(−1.770)	(−2.562)
Privacy	0.001	0.009	−0.019	−0.014*	−0.006	−0.031*
	(0.101)	(0.833)	(−1.292)	(−1.695)	(−0.506)	(−1.973)
Intercept	−0.003	0.015	0.035	−0.002	0.016	0.038
	(−0.251)	(0.814)	(1.402)	(−0.112)	(0.844)	(1.396)
Sample size	120	113	105	120	113	105
R^2	0.074	0.144	0.104	0.101	0.156	0.126
Adj. R^2	0.008	0.078	0.030	0.036	0.091	0.053
F-value	1.116	2.182**	1.395	1.559	2.406**	1.729

Note(s): 1. t-values are in parenthesis
2. Impact Scope: 1 if the number of people affected by an event is large, and 0 if the number of people affected by an event is small; I&D (Injuries and Deaths): 1 if there was someone injured or killed in the events, 0 otherwise; Industry Type: 1 if the firm was a technology firm, 0 otherwise; Firm Size: 1 for large firm, 0 for small firm; Event Year: 1 if the event took place after year 2019, 0 otherwise; Event Type: base group is fairness service failure event, accuracy, safety and privacy service failure events were dummy variables
*$p < 0.10$, **$p < 0.05$, ***$p < 0.01$
Source(s): Table created by authors

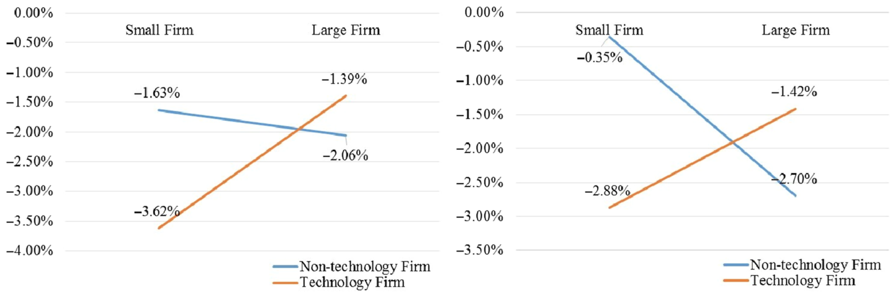
Table 16. Interaction effect of firm size and industry type

	[0,1]	[0,1]	[0,3]	[−1,3]	[0,5]	[−1,5]
Impact scope	−0.003	0.002	−0.013	−0.007	−0.017	−0.013
	(−0.672)	(0.269)	(−1.553)	(−0.874)	(−1.489)	(−1.070)
I&D (injuries and deaths)	−0.000	−0.000	0.007	0.006	0.017	0.017
	(−0.016)	(−0.019)	(0.481)	(0.451)	(0.944)	(0.837)
Industry type	−0.004	−0.004	−0.027**	−0.027**	−0.036**	−0.036**
	(−0.542)	(−0.415)	(−2.176)	(−2.101)	(−2.212)	(−2.039)
Firm size	0.005	0.013	−0.009	−0.002	−0.030*	−0.022
	(0.650)	(1.217)	(−0.677)	(−0.115)	(−1.672)	(−1.160)
Event year	−0.006	−0.004	−0.012	−0.010	−0.005	−0.004
	(−1.324)	(−0.648)	(−1.548)	(−1.247)	(−0.458)	(−0.316)
Accuracy	−0.004	−0.014*	−0.018*	−0.028***	−0.039***	−0.048***
	(−0.681)	(−1.776)	(−1.840)	(−2.688)	(−2.903)	(−3.273)
Safety	−0.006	−0.013	−0.018	−0.025*	−0.043**	−0.050**
	(−0.715)	(−1.228)	(−1.298)	(−1.744)	(−2.363)	(−2.524)
Privacy	0.001	−0.014*	0.009	−0.006	−0.020	−0.031**
	(0.089)	(−1.684)	(0.815)	(−0.527)	(−1.377)	(−2.037)
Firm Size*Industry Type	0.004	−0.001	0.028*	0.022	0.040*	0.034
	(0.420)	(−0.110)	(1.695)	(1.316)	(1.842)	(1.436)
Constant	−0.001	−0.002	0.025	0.024	0.047*	0.048*
	(−0.098)	(−0.141)	(1.323)	(1.225)	(1.828)	(1.713)
Observations	120	120	113	113	105	105
R^2	0.076	0.101	0.167	0.170	0.135	0.145

Note(s): *T*-statistics in parentheses
***$p < 0.01$, **$p < 0.05$, *$p < 0.1$
Source(s): Table created by authors

service failure events had a stronger negative impact on firm value for technology firms than non-technology firms. For large firms, AI service failure events had a stronger negative impact on firm value for non-technology firms than for technology ones (see Figure 3). In addition, the coefficients of industry type were significantly negative during the [0,3], [0,5], [−1,3], and [−1,5] windows in the regression model with interaction terms. These results suggested that the limited significant effect of industry type in previous regression might be due to the interaction effect offsetting or canceling out the main effect. This analysis provided more empirical evidence to support the heterogeneous effects of industry type.

Second, this study investigates how different types of AI service failures interact with firm size. As shown in Table 17, the coefficient of the interaction term between firm size and



Source(s): Figure created by authors

Figure 3. Mean CARs for [0, 3] and [0, 5] windows

Table 17. Interaction effects of firm size and event type

	[0,1]	[−1,1]	[0,3]	[−1,3]	[0,5]	[−1,5]
Impact scope	−0.003	0.003	−0.011	−0.006	−0.015	−0.012
	(−0.523)	(0.414)	(−1.354)	(−0.683)	(−1.340)	(−0.959)
I&D (injuries and deaths)	−0.001	−0.002	−0.001	−0.003	0.006	0.003
	(−0.129)	(−0.223)	(−0.061)	(−0.205)	(0.281)	(0.136)
Industry type	−0.002	−0.006	−0.012	−0.015*	−0.016	−0.021*
	(−0.439)	(−0.976)	(−1.387)	(−1.825)	(−1.431)	(−1.736)
Firm size	0.001	−0.012	0.009	−0.005	−0.002	−0.019
	(0.144)	(−1.068)	(0.586)	(−0.324)	(−0.080)	(−0.909)
Event year	−0.007	−0.006	−0.013	−0.012	−0.007	−0.008
	(−1.437)	(−0.984)	(−1.571)	(−1.462)	(−0.585)	(−0.637)
Accuracy	−0.007	−0.031***	−0.023	−0.047***	−0.035*	−0.060***
	(−0.813)	(−2.838)	(−1.528)	(−3.135)	(−1.820)	(−2.941)
Safety	−0.008	−0.021	−0.012	−0.025	−0.032	−0.046
	(−0.603)	(−1.303)	(−0.522)	(−1.033)	(−1.024)	(−1.390)
Privacy	−0.011	−0.041***	0.011	−0.020	−0.030	−0.061***
	(−1.110)	(−3.423)	(0.683)	(−1.208)	(−1.433)	(−2.755)
Firm Size*Accuracy	0.005	0.029**	0.009	0.035*	−0.009	0.020
	(0.442)	(2.090)	(0.484)	(1.807)	(−0.355)	(0.741)
Firm Size*Safety	0.004	0.016	−0.007	0.005	−0.017	−0.002
	(0.301)	(0.950)	(−0.265)	(0.203)	(−0.507)	(−0.051)
Firm Size*privacy	0.020	0.048***	−0.005	0.023	0.023	0.059*
	(1.616)	(3.007)	(−0.222)	(1.080)	(0.803)	(1.924)
Constant	0.002	0.016	0.016	0.030	0.035	0.056*
	(0.179)	(1.033)	(0.739)	(1.408)	(1.215)	(1.798)
Observations	120	120	113	113	105	105
R-squared	0.099	0.176	0.150	0.187	0.122	0.167

Note(s): *T*-statistics in parentheses
***$p < 0.01$, **$p < 0.05$, *$p < 0.1$
Source(s): Table created by authors

accuracy service failure event was significantly positive in the event windows of [−1,1] and [−1,3], and the coefficient of interaction term between firm size and privacy service failure event was significantly positive in the event windows of [−1,1] and [−1,5]. These results indicated a significant interaction effect between event type and firm size. The negative impacts of privacy service failure events on firm value were significantly stronger for small firms than large firms. The negative impacts of accuracy service failure events on firm value were significantly stronger for small firms than for large firms.

6. Discussion

6.1 Findings

This study investigated the stock market's reactions to AI service failure events using the event study method. The results revealed several interesting findings.

First, our results showed that AI service failure events negatively impact the market value of firms. This finding aligns with those of previous security breach event studies that revealed negative technology-related events have a negative effect on firms' market value (Ebrahimi and Eshghi, 2022; Spanos and Angelis, 2016). In addition, the magnitude of the effect generated by AI service failure events is larger than that of previous security breach events, emphasizing the importance of AI service failure events and calling for attention on them.

Second, we examined how contextual factors moderate the negative relationship between AI service failure events and market response. The results indicated that technology firms experience more share price declines in the short term, due to AI service failure events, than non-technology firms. This result is reasonable since users are more intolerant of service failure events occurring in technology companies, which are supposed to be better at avoiding technical problems. Consistent with previous studies on negative signals (Bharadwaj and Keil, 2001; Ebrahimi and Eshghi, 2022), we also found that small firms experienced more negative reactions to AI service failure events than large firms due to their insufficient capabilities to cope with negative events. In addition, our results suggest that AI service failure events in more recent years have more intensively negative impacts than those in more distant years. As predicted (Cavusoglu *et al.*, 2004), AI service failure events are gradually being understood and taken seriously by more investors, thus causing an increase in their negative impacts. It should be noted that the comparison of the subsamples from 2010 to 2014 and from 2015 to 2019 showed an unexpected result where in the [0, 1] event window events that occurred before 2014 had a stronger negative impact. One probable reason was that the duration and pattern of the events' impact changed. As an emerging event, AI service failure events in earlier years (before 2014) had a short but sharp impact on the stock market, resulting in a significant decline in firm value during the [0,1] window. In the later period, the duration of the impact of service failure events became longer, and the size of the impact increased with time, resulting in a more significant decrease in firm value during the [0,3], [0,5] windows. Thus, the impact of events in earlier years (before 2014) was more pronounced at an early stage, while the impact of AI service failure events was relatively delayed in recent years (2015–2019).

Third, we identified stock market reactions to various types of AI service failure events. The results show that accuracy service failure events and safety service failure events have significantly more negative impacts on firm value, privacy service failure events have relatively weak negative impacts on firm value, while the effect of fairness service failure events on firm value is not significant. Both the impact magnitude and the impact longevity of accuracy and safety service failure events on firms' market value are larger than those of privacy service failure events. Meanwhile, the impact magnitude of accuracy service failure events on firm value is larger than those of safety service failure events. These results revealed that investors focus the most on accuracy issues, the second on safety issues, then privacy issues, but do not perceive AI fairness issues as concerns.

Fourth, we investigated the significant interaction effects between independent variables. Considering the interaction of industry type and firm size, AI service failure events have a stronger negative impact on market value in technology firms when the firm size is small, while the impact on firm value is more negative for non-technology firms when the firm size is large. The possible reason for this result is that large technology firms are involved in a wider variety of technologies (Cavusoglu *et al.*, 2004), with AI technology as only one aspect. Even if AI services failure occurs, it does not mean that the firm's services suffer a catastrophic blow, thus weakening its negative impact. However, small technology firms involve relatively fewer technology types or even just one, and the importance of AI technology in small technology firms is higher, thus they are more sensitive to AI service failures. In addition, the interaction effects between event type and firm size is significant. The negative impacts of accuracy and privacy service failure events on firm value are stronger for small firms than for large firms. Hence, certain types of AI service failure events have a higher importance for specific firm types.

6.2 Theoretical contributions

Our research has the following theoretical contributions. First, this study extends the application of signaling theory to the emerging context of AI service failure in firms. Previous studies mainly focused on positive signals in the stock market (Liu *et al.*, 2024; Rahman *et al.*, 2018; Tao-Schuchardt *et al.*, 2023), while ignoring the negative aspects. Our research reveals that AI service failure events have a negative impact on firm value. It provides timely and objective empirical evidence that AI service failure events can be seen as a negative signal to stakeholders on firm value. The current research enriches the understanding of the impacts of such signals by focusing on firm-level dependent variables rather than consumer-level ones. To the best of our knowledge, our research is an early attempt to evaluate and explain how the stock market responds to AI service failure events using the signaling theory.

Second, this study advances the understanding of AI service failure events' impacts by examining certain contextual factors, namely, industry type, firm size, and event year. Against the backdrop that the effects of contextual influencing factors are mixed (Bharadwaj and Keil, 2001; Cavusoglu *et al.*, 2004; Ebrahimi and Eshghi, 2022; Yayla and Hu, 2011), our study suggests that there is a stronger drop in market value for technology firms than non-technology firms, for small firms than large firms, and in recent years than earlier years. Our study offers new empirical evidence that market reactions differ depending on industry type, firm size, and event year in the context of AI service failures. Furthermore, we find that the interaction between firm size and industry type affects market reactions to AI service failure events. These findings bring new insights to the research on AI service failure events by including a comprehensive and diverse set of contextual factors and their interaction effects.

Third, this study contributes to the literature on service failure by proposing a novel classification of accuracy, safety, fairness, and privacy service failures. Unlike previous studies that divided service failures into two categories (Choi *et al.*, 2021; Song *et al.*, 2023), this study uses a classification with four categories that focus on the aspects related to service outcomes. This classification is more nuanced and captures key dimensions of customer requirements. It provides a useful perspective to evaluate and explain the potential costs of AI service failure events and promotes a better understanding of service failure by identifying the effects of different event types.

Fourth, this study provides useful insights for firm resource management research by examining the impacts of AI service failure event types. Extant work on firm behavior after negative events has largely focused on service recovery behavior, such as apologies and compensation (Gwebu *et al.*, 2018). There is a lack of research on what aspects companies should invest to mitigate negative events from the perspective of resource allocation. Our results reveal that the stock market reacts differently to different types of AI service failure events and that accuracy service failures are viewed by investors as the most detrimental, followed by safety service failure events. The short-term market reactions of investors or stakeholders to privacy and fairness issues are relatively weak or even nonsignificant. Moreover, we found that for small firms and accuracy and privacy service failure events generate stronger negative impacts on firm value. Hence, this research broadens our understanding of the content and type of service recovery expected by users and sheds light on how firms can allocate resources to prevent the negative impacts of certain types of service failures to achieve better market reactions.

6.3 Practical implications

This study offers important insights and valuable guidance to managers and developers. First, our results highlight the importance of AI service failure events in the stock market. The decline in firm value after AI service failures and the increasing magnitude of the decline over time indicate that AI service failure events have been seen as a negative quality signal and attracted more and more attention from shareholders in recent years. Companies without AI services should carefully make their AI adoption decisions by assessing the likelihood and potential loss of AI service failures so that they can cope with the negative effects. Firms with AI services should pay close attention to media reports of AI service failure events and respond in a timely manner. Meanwhile, firms can strengthen the publicity of AI service capabilities and quality to eliminate the impact of negative signals.

Second, the heterogeneity in the impacts of firm characteristics reveals how companies should prepare for AI service failure events to minimize their negative effects. The moderating effect of firm size highlights that large firms have a stronger ability to cope with risks. Therefore, managers can improve their firm's risk-taking capability by establishing systematic evaluation methods and coping strategies for AI service failure events. Constructing a reliable brand image can also help reduce negative market impact. The interaction effect of industry type and firm size also underscores the importance of technological diversity. Firms should expand their technology types and reduce their reliance on AI technology, thus reducing their sensitivity to AI service failure events.

Third, this study proposed a useful resource allocation strategy by highlighting areas businesses should improve to match user requirements. Our study reveals the comparison results on the impacts of event type and the interaction effect between event type and firm size. Such results derive from the heterogeneity of user requirements in different scenarios. Therefore, it is necessary to conduct in-depth research on user needs for AI services in different contexts and provide matching AI service designs. When firms' resources are limited, they can prioritize resource allocation that strengthens AI capabilities to avoid accuracy issues and meet the most basic needs of users. Then, the second need of safety should be satisfied for AI developers to optimize the effective mechanism that can protect users' lives and property. Moreover, AI service designers should strengthen security systems to prevent data breaches and enhance user data rights to avoid data abuse, especially in small firms. It should be noted that although fairness issues may not show a great direct impact on firm value within the scope of this study, they may still be important since their impacts may go beyond immediate financial metrics. For example, the service failure events on fairness may bring ethical issues and have great harms on users' trust and public attitudes toward AI. Therefore, developers should also consider the ethical design of fairness and create suitable methods to evaluate the non-economic effects of such issues. However, the lack of market reaction to such ethical issues is likely to make companies lose the motivation to optimize services and ignore the aspect of fairness issues. More powerful regulatory measures should be proposed by the government institutes to protect users' rights.

6.4 Limitations and future research

Similar to other event studies, this study has some limitations. First, we identified the event date manually through public media, but there is usually a gap between the public disclosure date and the actual AI service failure event date. It is difficult to determine whether the event was leaked out to investors and individuals before disclosure through public media. Alternative data sources can be used in future research to obtain a more exact event date and to capture market reactions more accurately. Second, this study only examines the negative impacts of AI service failure events on firm value in a relatively short period. It is worthwhile for future studies to explore the long-term effects of AI service failure events and their effects on other outcome variables such as sales, revenue, and other financial indices. Third, the sample contains service failure events that happened to various AI applications, such as recommendation algorithms, face recognition, autopilot, and chatbots. Future studies can focus on specific AI application service failures and identify their impacts on firm value to eliminate the confounding effect of AI application type. Fourth, this study only uses four aspects related to service outcomes as the types of AI service failure, ignoring many other ethical issues related to the service interaction process. Further studies should include additional dimensions of AI service failures, such as explainability, accountability, and machine autonomy to further examine the impact of the type of service failure.

7. Conclusion

In this study, we investigated the impacts of AI service failure events on firm stock value using a sample of 120 AI service failure events in publicly traded U.S. firms. Based on the empirical analysis, we found a negative effect of AI service failure events on firm stock value. Specifically, our results revealed that investors react more negatively to AI service failure events in technology firms than non-technology firms, in small firms than large firms, and in recent years than earlier years. There is also an interaction effect between firm size and industry type on the market reactions of AI service failure events. In addition, the impacts of AI service failure events on firm value varied across the event types. Accuracy service failure events resulted in a more pronounced reduction in stock value compared with safety service failure events, privacy service failure events brought a relatively weak reduction in firm value, while the effect of fairness service failure events on firm value was not significant. Moreover, our results indicated that the stock price declined more dramatically due to accuracy and privacy AI service failure events in small firms compared with large firms. Overall, this study offers empirical evidence on negative market reactions caused by AI service failure events and contributes to our comprehensive understanding of the effect of AI service failure events, thus providing useful guidance to managers and AI developers for service improvement and strategy development.

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(The Appendix follows overleaf)

Appendix

Table A1. Descriptive information of sample firms

Company name	Stock code	Market	Date of listing	Number of employees	Industry type
Altaba Inc	AABA	NASDAQ	1996/4/12	8,500	Non-Technology Firm
Apple Inc	AAPL	NASDAQ	1980/12/12	161,000	Technology Firm
Adobe Inc	ADBE	NASDAQ	1986/8/15	29,945	Technology Firm
Amazon.Com, Inc	AMZN	NASDAQ	1997/5/15	1,525,000	Technology Firm
The Arena Group Holdings, Inc	AREN	NYSE	2022/2/9	441	Technology Firm
Activision Blizzard, Inc	ATVI	NASDAQ	1993/10/22	13,000	Technology Firm
The Boeing Company	BA	NYSE	1934/9/5	171,000	Non-Technology Firm
Alibaba Group Holding Limited	BABA	NYSE	2014/9/19	224,955	Technology Firm
Bilibili Inc	BILI	NASDAQ	2018/3/28	8,801	Technology Firm
Buzzfeed, Inc	BZFD	NASDAQ	2021/3/5	735	Technology Firm
Cass Information Systems, Inc	CASS	NASDAQ	1996/7/1	1,061	Non-Technology Firm
The Cigna Group	CI	NYSE	2018/12/21	72,500	Non-Technology Firm
Zw Data Action Technologies Inc	CNET	NASDAQ	2010/9/14	64	Technology Firm
Cisco Systems, Inc	CSCO	NASDAQ	1990/2/16	84,900	Technology Firm
Cvs Health Corporation	CVS	NYSE	1952/10/1	300,000	Non-Technology Firm
Denny'S Corporation	DENN	NASDAQ	2005/5/10	3,500	Non-Technology Firm
The Walt Disney Company	DIS	NYSE	1957/11/12	225,000	Non-Technology Firm
The Estee Lauder Companies Inc	EL	NYSE	1995/11/16	62,000	Non-Technology Firm
Elbit Systems Ltd	ESLT	NASDAQ	1996/11/29	18,984	Technology Firm
Evolv Technologies Holdings, Inc	EVLV	NASDAQ	2020/9/22	293	Technology Firm
Exponent, Inc	EXPO	NASDAQ	1990/8/17	1,320	Technology Firm
Ford Motor Company	F	NYSE	1956/1/17	177,000	Non-Technology Firm
Gannett Co., Inc	GCI	NYSE	2014/2/4	11,200	Technology Firm
Ge Aerospace	GE	NYSE	1892-06-23	125,000	Technology Firm
General Motors Company	GM	NYSE	2010/11/18	163,000	Non-Technology Firm
Alphabet Inc	GOOGL	NASDAQ	2004/8/19	182,502	Technology Firm
The Goldman Sachs Group, Inc	GS	NYSE	1999/5/4	45,300	Non-Technology Firm
Hasbro, Inc	HAS	NASDAQ	1978/1/6	5,502	Non-Technology Firm
The Home Depot, Inc	HD	NYSE	1981/9/1	463,100	Non-Technology Firm
Honda Motor Co., Ltd	HMC	NYSE	1977/2/11	197,039	Non-Technology Firm
Hp Inc	HPQ	NYSE	1957/11/6	58,000	Technology Firm

(*continued*)

Table A1. Continued

Company name	Stock code	Market	Date of listing	Number of employees	Industry type
Hsbc Holdings Plc	HSBC	NYSE	1999/7/16	220,861	Non-Technology Firm
Humana Inc	HUM	NYSE	1971/5/18	67,600	Non-Technology Firm
International Business Machines Corporation	IBM	NYSE	1915/11/11	282,200	Technology Firm
Intel Corporation	INTC	NASDAQ	1971/10/13	124,800	Technology Firm
Irobot Corporation	IRBT	NASDAQ	2005/11/9	1,113	Technology Firm
The Kroger Co	KR	NYSE	1928/1/26	414,000	Non-Technology Firm
L3harris Technologies, Inc	LHX	NYSE	1955/7/13	50,000	Technology Firm
Southwest Airlines Co	LUV	NYSE	1971/6/8	74,806	Non-Technology Firm
Lyft, Inc	LYFT	NASDAQ	2019/3/29	2,945	Technology Firm
Macy'S, Inc	M	NYSE	1992/2/7	85,581	Non-Technology Firm
Mcdonald'S Corporation	MCD	NYSE	1965/4/21	150,000	Non-Technology Firm
Meta Platforms, Inc	META	NASDAQ	2012/5/18	67,317	Technology Firm
Hello Group Inc	MOMO	NASDAQ	2014/12/11	1,382	Technology Firm
Microsoft Corporation	MSFT	NASDAQ	1986/3/13	221,000	Technology Firm
Match Group, Inc	MTCH	NASDAQ	2020/7/1	2,600	Technology Firm
Netflix, Inc	NFLX	NASDAQ	2002/5/23	13,000	Technology Firm
Nio Inc	NIO	NYSE	2018/9/12	32,820	Technology Firm
Nuance Communications, Inc	NUAN	NASDAQ	1995/12/11	6,900	Technology Firm
Nvidia Corporation	NVDA	NASDAQ	1999/1/22	29,600	Technology Firm
Oracle Corporation	ORCL	NYSE	2013/7/15	164,000	Technology Firm
Ryanair Holdings Plc	RYAAY	NASDAQ	1997/6/5	22,261	Non-Technology Firm
Starbucks Corporation	SBUX	NASDAQ	1992/6/26	381,000	Non-Technology Firm
Snap Inc	SNAP	NYSE	2017/3/2	5,289	Technology Firm
Spotify Technology S.A.	SPOT	NYSE	2018/4/3	9,123	Technology Firm
Soundthinking, Inc	SSTI	NASDAQ	2017/6/7	312	Technology Firm
Target Corporation	TGT	NYSE	1969/9/8	415,000	Non-Technology Firm
Toyota Motor Corporation	TM	NYSE	1999/9/29	375,235	Non-Technology Firm
Transunion	TRU	NYSE	2015/6/25	13,200	Non-Technology Firm
Tesla, Inc	TSLA	NASDAQ	2010/6/29	140,473	Technology Firm
Twitter, Inc	TWTR	NYSE	2013/11/7	7,500	Technology Firm
Uber Technologies, Inc	UBER	NYSE	2019/5/10	30,400	Technology Firm
Unitedhealth Group Incorporated	UNH	NYSE	1984/10/19	440,000	Non-Technology Firm
Upstart Holdings, Inc	UPST	NASDAQ	2020/12/16	1,388	Technology Firm
Weibo Corporation	WB	NASDAQ	2014/4/17	5,268	Technology Firm
Workday, Inc	WDAY	NASDAQ	2017/9/20	18,800	Technology Firm
The Wendy'S Company	WEN	NASDAQ	2011/12/27	15,300	Non-Technology Firm
Walmart Inc	WMT	NYSE	1970/10/1	2,100,000	Non-Technology Firm

(*continued*)

Table A1. Continued

Company name	Stock code	Market	Date of listing	Number of employees	Industry type
Ww International, Inc	WW	NASDAQ	2018/10/15	4,850	Non-Technology Firm
Xpeng Inc	XPEV	NYSE	2020/8/27	13,550	Non-Technology Firm
Yandex N.V.	YNDX	NASDAQ	2011/5/24	26,361	Technology Firm
Yum! Brands, Inc	YUM	NYSE	1997/10/7	35,000	Non-Technology Firm
Zillow Group, Inc	Z	NASDAQ	2015/8/17	6,263	Non-Technology Firm
Zhihu Inc	ZH	NYSE	2021/3/26	2,731	Technology Firm
Zoom Video Communications, Inc	ZM	NASDAQ	2019/4/18	7,420	Technology Firm

Source(s): Table created by authors

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